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Roll No. : 317 Sub : Computer Networks

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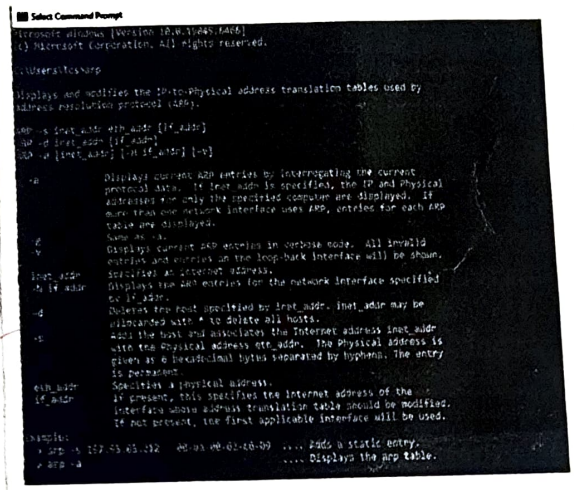
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Experiment-1
Date: 25/07/26 Study of various Network commands used in Linux and windows:-

Windows :-

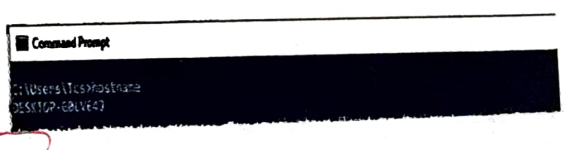
1) arp (Address Resolution protocol)

Output :-



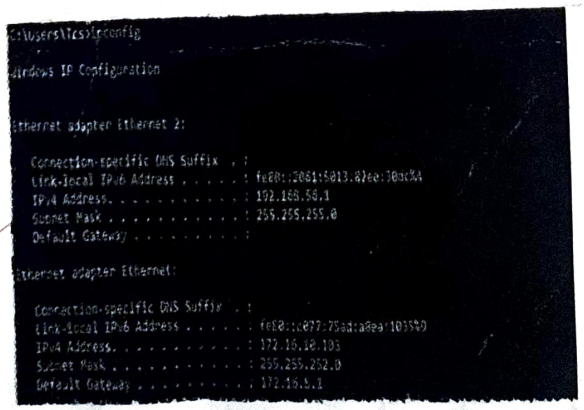
2) hostname :

Output :-



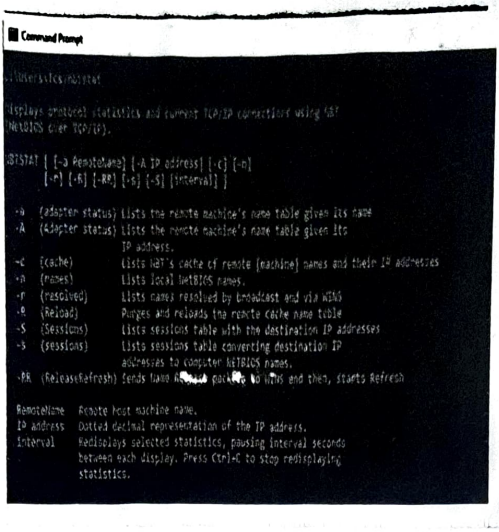
3) ipconfig /all :-

Output :-



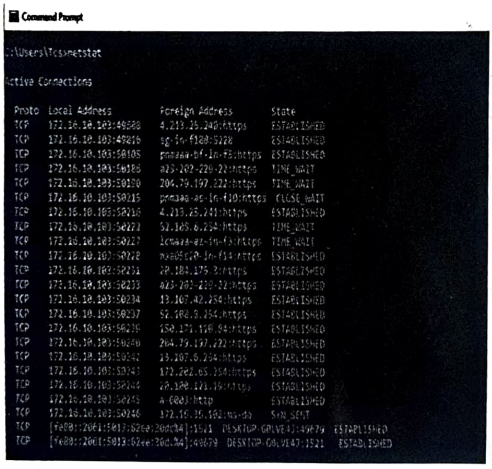
4) nbtstat :-

Output :-



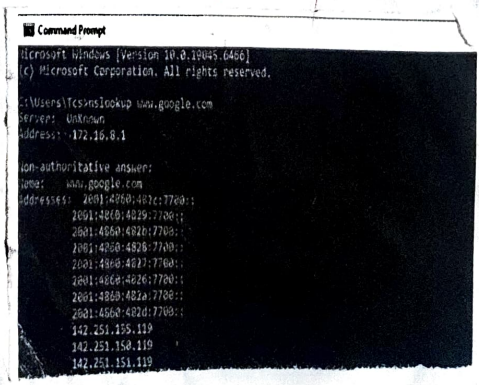
5) netstat :

Output :-



6) nslookup :

Output :-



7) pathping :-

Output :-

Command Prompt

```
C:\Users\Tcs>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
  -g host-list     Loose source route along host-list.
  -h maximum_hops  Maximum number of hops to search for target.
  -i address       Use the specified source address.
  -n               Do not resolve addresses to hostnames.
  -p period        Wait period milliseconds between pings.
  -q num_queries   Number of queries per hop.
  -w timeout       Wait timeout milliseconds for each reply.
  -4               Force using IPv4.
  -6               Force using IPv6.
```

8) ping :-

Output :-

Command Prompt

```
C:\Users\Tcs>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

9) Route :-

Output :-

Command Prompt

```
C:\Users\Tcs>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

    -f           Clears the routing tables of all gateway entries.  If this is
                 used in conjunction with one of the commands, the tables are
                 cleared prior to running the command.

    -p           When used with the ADD command, makes a route persistent across
                 boots of the system.  By default, routes are not preserved
                 when the system is restarted.  Ignored for all other commands,
                 which always affect the appropriate persistent routes.

    -4           Force using IPv4.

    -6           Force using IPv6.

    command      One of these:
                   PRINT     Prints a route
                   ADD       Adds a route
                   DELETE    Deletes a route
                   CHANGE    Modifies an existing route
    destination  Specifies the host.
    MASK         Specifies that the next parameter is the 'netmask' value.
    netmask      Specifies a subnet mask value for this route entry.
                 If not specified, it defaults to 255.255.255.255.
    gateway      Specifies gateway.
    interface    the interface number for the specified route.
    METRIC       specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.
```

Command Prompt

```
If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a '*' or '?', it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any single character. Examples: 157.*.1, 157.*, 127.*, *224*

Pattern match is only allowed in PRINT command.

Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example: route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT 157*       .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
                destination^  ^mask      ^gateway  ^metric
                                                   ^interface
    If IF is not given, it tries to find the best interface for a given
    gateway.
    > route ADD 3ffe::0 MASK ffff:: 3ffe::1
    or
    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

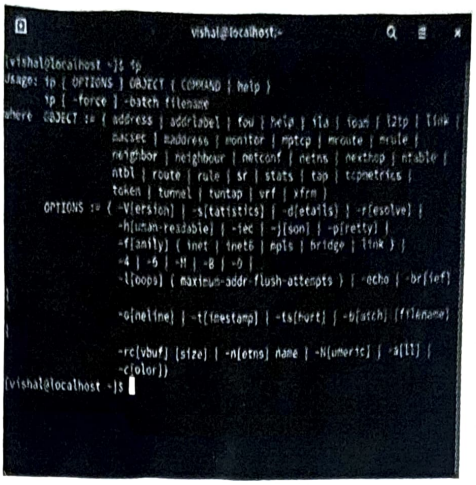
    PRINT is used to print a route, also in combination with wildcards.

    > route PRINT
    > route PRINT 157*
```

Linux :-

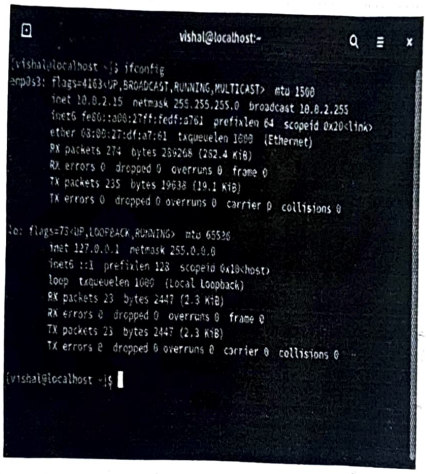
1) ip

Output :-



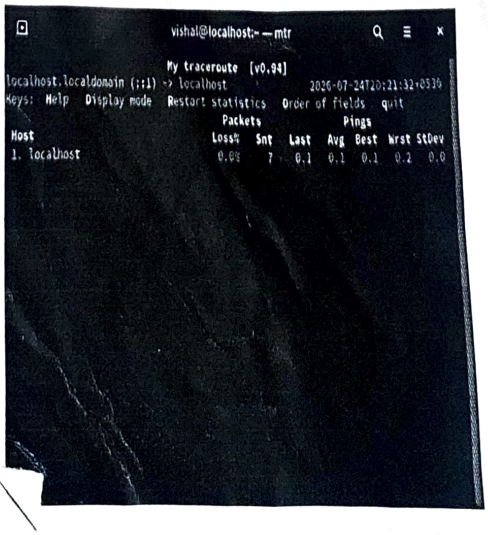
2) ipconfig

Output :-



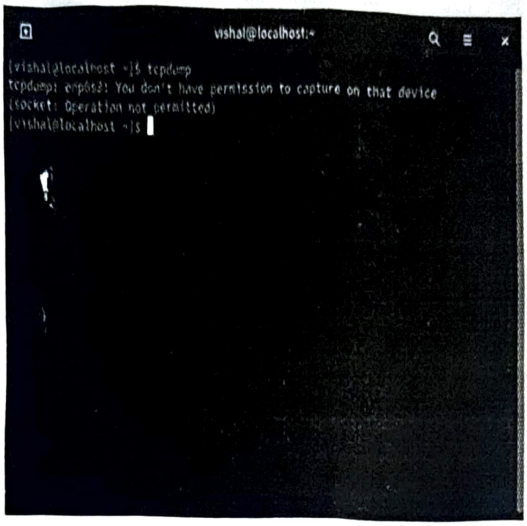
3) mtr

Output :-



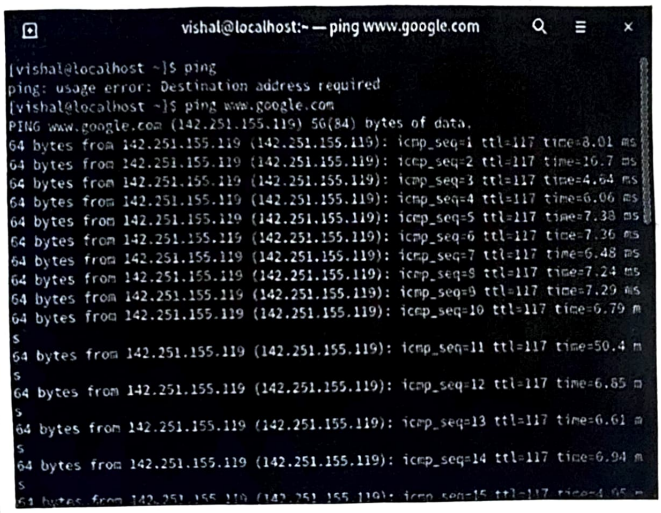
4) tcpdump

Output:-



5) Ping

Output:-



Result:-

Thus Network commands used in Linux & Windows are executed successfully.